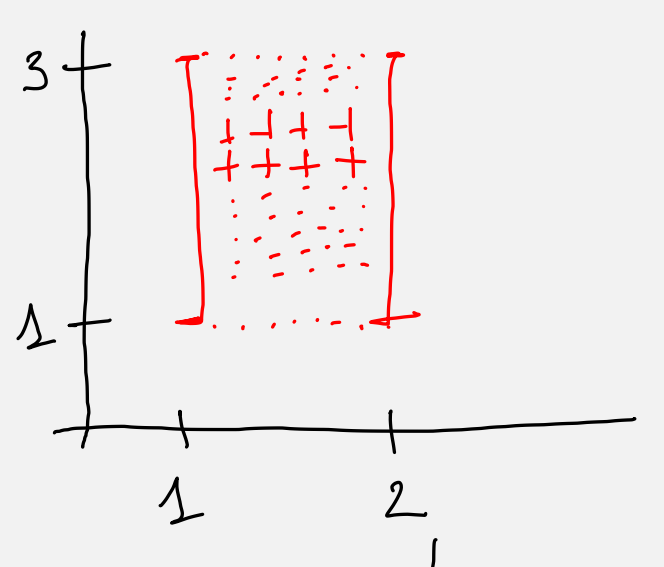
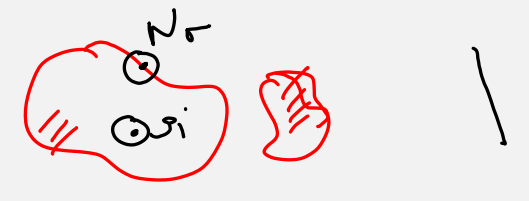
$B = A_1 \cap \mathbb{Q}^2$



$\text{Int}(B) := \{p \in \mathbb{R}^2 \mid \exists \varepsilon > 0, B(p,\varepsilon) \subset B\}$



$p \notin B \Rightarrow p \notin \text{Int}B$

$p \in B$? $\forall \varepsilon > 0$, $B(p,\varepsilon)$.

$p + \left(\sqrt{2} \cdot \frac{1}{N}, 0\right)$

$\|p - (p + (\sqrt{2} \times \frac{1}{N}, 0))\|_2 = \sqrt{(p_x - p_x - \sqrt{2}\frac{1}{N})^2 + (p_y - p_y)^2}$

$= |\sqrt{2} \times \frac{1}{N}| < \varepsilon$, $\sqrt{2}/N \in \mathbb{R} \setminus \mathbb{Q}$ ⊗

$\sqrt{2}/N = a/b \Rightarrow \sqrt{2} = \frac{Na}{b} \in \mathbb{Q}$

$p + (\sqrt{2}/N)\vec{e_1} \in B(p,\varepsilon)$

$\notin B \cap \mathbb{Q}^2$

$p_x + \frac{\sqrt{2}}{N} = \frac{a}{b} \Rightarrow \sqrt{2} = N(\frac{a}{b} - p_x) \in \mathbb{Q}$ ⊗

o sea: $\text{Int}(B) = \emptyset$.

$\text{Adh}(B)$? $\text{Fr}(B)$?

$\text{Adh}(B) = \{p \in \mathbb{R}^2 \mid \forall \varepsilon > 0, B(p,\varepsilon) \cap B \neq \emptyset\}$.

$B \subset \text{Adh}(B)$, $p \in B$, $B(p,\varepsilon) \cap B \ni p$

$\mathbb{Q}$ denso en $\mathbb{R}$: $\text{Adh}(\mathbb{Q}) = \mathbb{R}$

$\mathbb{Q} \cap I$, $\text{Adh}(\mathbb{Q} \cap I) = I$.

$A_1 = [\underset{\overset{\psi}{x}}{1,2}] \times (\underset{\overset{\psi}{y}}{1,3})$ $\ni (x,y)$

$\text{Adh}(\mathbb{Q}^2 \cap A_1) = \text{Adh}(A_1)$

$\text{Adh}(B) = [1,2] \times [1,3]$

$\forall p \in \text{Adh}(A)$, $\exists (p_n) \in (\mathbb{Q}^2 \cap A_1)^{\mathbb{N}}$ tq

$\|p_n - p\| \xrightarrow[n\to+\infty]{} 0$

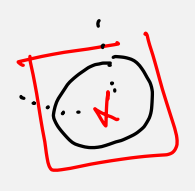
$x \in \mathbb{R}$, $\exists x_n \in \mathbb{Q}^{\mathbb{N}}$ tq $x_n \xrightarrow[n\to+\infty]{} x$.

p_x $\exists (x_n)$ tq $x_n \to p_x$

p_y, (y_n) tq $y_n \to p_y$ con:

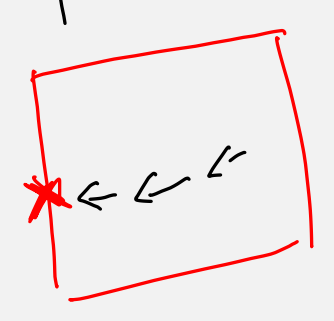
Si. $|x_n - p_x| \to 0$ entonces $\|(p_x,p_y) - (x_n,y_n)\| \xrightarrow[n\to+\infty]{} 0$ Prop

$|y_n - p_y| \to 0$

Si  no el pinen ahno tq $\forall n$, $(x_n,y_n) \in A$, $(p_{\varphi(n)}) = (x_{n+n_0}, y_{n+n_0})$

Como $\forall n, (x_n,y_n) \in \mathbb{Q}^2$, entonces $(p_n) \in (A \cap \mathbb{Q}^2)^{\mathbb{N}} = (B)^{\mathbb{N}}$

$p \in \text{Adh}(B)$.



Prueba Prop: Supongo que $|x_n - x| \to 0$, $|y_n - y| \to 0$.

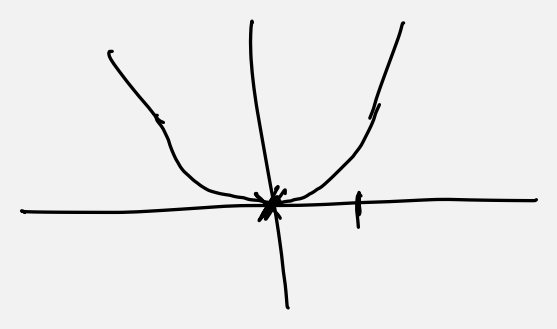
$\|(x,y) - (x_n,y_n)\|_2 \xrightarrow[n\to+\infty]{} 0 \Leftrightarrow \|(x,y) - (x_n,y_n)\|_2^2 \xrightarrow[n\to+\infty]{} 0$

Quiero probar: $(x - x_n)^2 + (y - y_n)^2 \xrightarrow[n\to\infty]{} 0$

$= |x - x_n|^2 + |y - y_n|^2$

Si $\forall \varepsilon > 0, \exists n_0 \in \mathbb{N} \mid n \geq n_0, \Rightarrow |x - x_n| < \varepsilon$ ⊕

$\Rightarrow n \geq n_0 \Rightarrow |x - x_n|^2 < \underset{\smile}{\varepsilon^2} < \varepsilon'$



⊕ $\forall \varepsilon' > 0, \exists n_0 \mid \forall n \geq n_0, |x - x_n|^2 < \varepsilon'$

Entonces $|x - x_n|^2 + |y - y_n|^2 \xrightarrow[n\to+\infty]{} 0$.

$\text{Adh}(B) = \text{Adh}(A) = [1,2] \times [1,3]$.

$\text{Fr}(B)$ ⓘ, $\text{Fr}(B) := \text{Adh}(B) \setminus \text{Int}(B)$

$\text{Adh}(B) = [1,2] \times [1,3]$

$\text{Int}(B) = \emptyset$

$\text{Fr}(B) = \text{Adh}(B) = [1,2] \times [1,3]$.

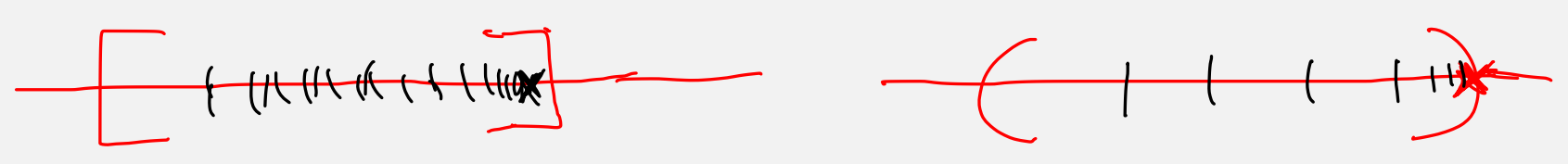
B Abierto ?

$\text{Int}B = \emptyset$, $B \neq \emptyset$

→ B no puede ser abierto.

Cerrado ? C es cerrado $\Leftrightarrow \forall (x_n) \in C^{\mathbb{N}}$ tq $x_n \to x$

tiene su límite $x \in C$.



$(0,1)$. $a_n = 1/n$

$1/n \xrightarrow[n\to+\infty]{} 0$, y $(a_n) \in (0,1)^{\mathbb{N}}$,

pero. $\text{Lim}\, a_n \notin (0,1)$.

C es cerrado $\Leftrightarrow \mathbb{R}^N \setminus C$ es abierto.

- Si supongo el criterio sequencial.

Sea $x \in \mathbb{R}^N \setminus C$, Si $\forall \varepsilon > 0$, $B(x,\varepsilon) \cap C \neq \emptyset$

Entonces $\varepsilon_n = 1/n$, como $\nearrow \neq \emptyset$, $x_n \in B(x,\varepsilon_n)$

y entonces $x_n \to x \Rightarrow x \in C$ absurdo ⊗

tiene que $\exists \varepsilon > 0$ tq $B(x,\varepsilon) \cap C = \emptyset$ o sea $x \in \text{Int}(\mathbb{R} \setminus C)$

o sea $\mathbb{R}^N \setminus C$ es abierto!

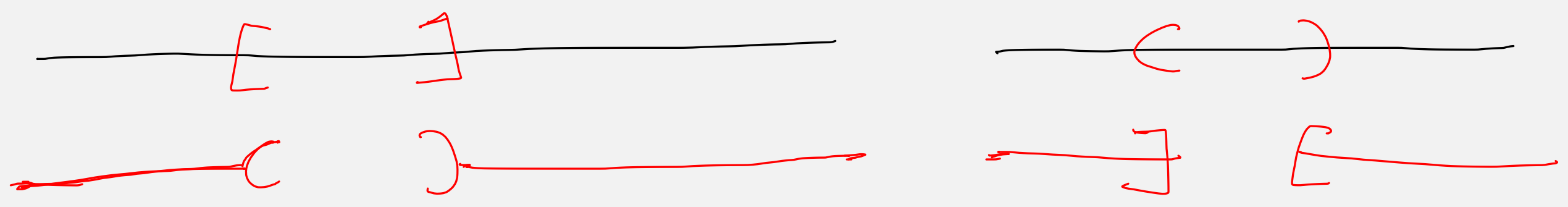
- Si $\mathbb{R}^N \setminus C$ es abierto $\Rightarrow$ Criterio sequencial.

Si $(x_n)_n \in C^{\mathbb{N}}$, y $x_n \to x$, si $x \in \mathbb{R}^N \setminus C$

no va haber $B(x,\varepsilon)$ tq $B(x,\varepsilon) \subset \mathbb{R}^N \setminus C$

porque va haber un $x_n \in B(x,\varepsilon) \cap C$, o sea $B(x,\varepsilon) \not\subset \mathbb{R}^N \setminus C$.

o sea $\mathbb{R}^N \setminus C$ no es abierto ⊗.



$(\sqrt{2} - 1, 2) \notin B$ $\sqrt{2} \simeq 1,4\ldots$, $x \mapsto \sqrt{x} < x$.

$\sqrt{1} = 1$, $\underset{\overset{\shortparallel}{1}}{\sqrt{1}} < \sqrt{2} < 2$

$x_n \to \sqrt{2} - 1$

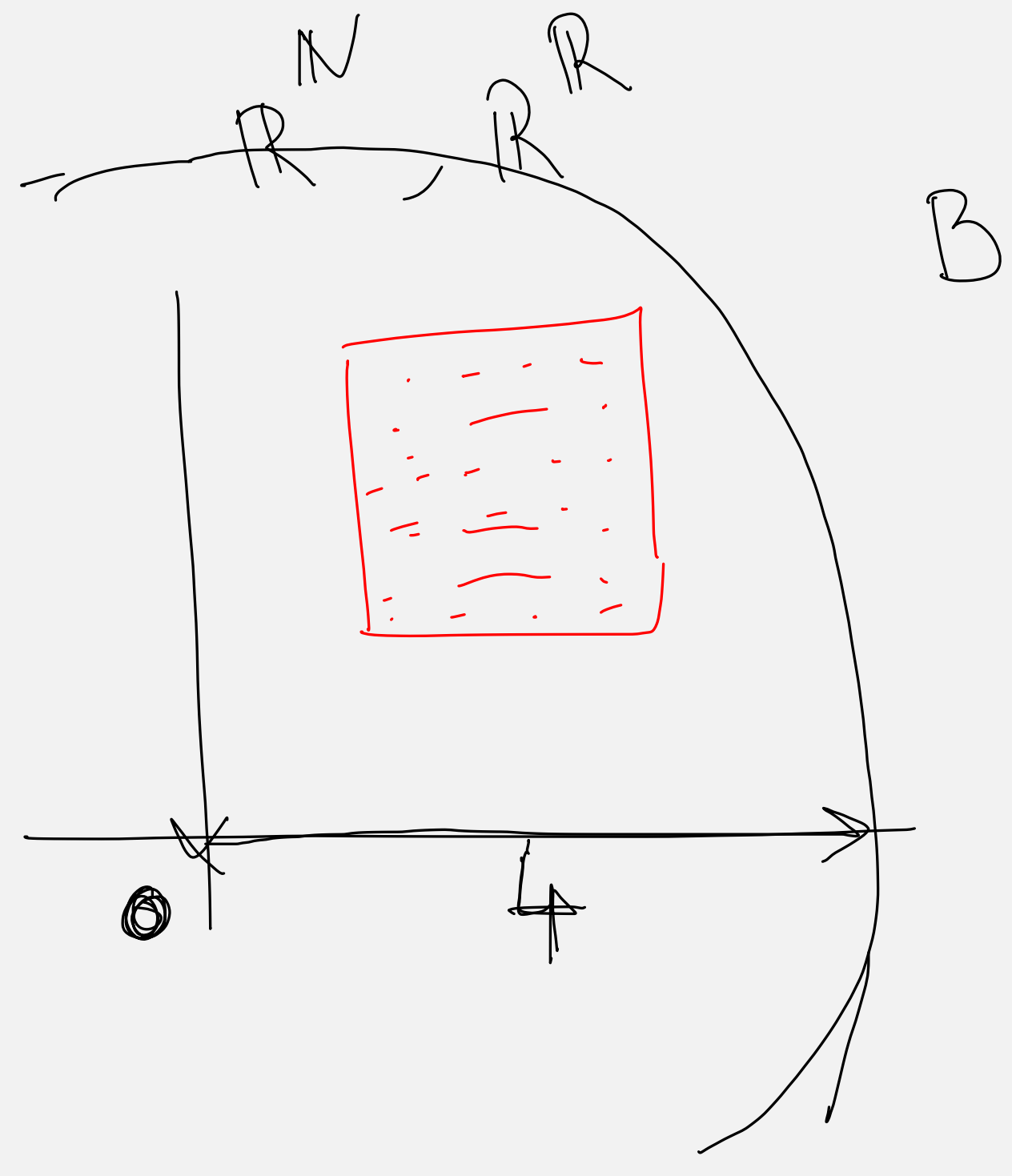
$\in \mathbb{Q}$, $(x_n)_n \in [1,2]^{\mathbb{N}}$.

$p_n = (x_n, 2) \xrightarrow[\|\cdot\|_2]{} (\sqrt{2} - 1, 2)$

$p_n \in B^{\mathbb{N}}$, pero $\text{Lim}\, p_n \notin B$. Entonces B no es cerrado.

B compacto $d(x,y) = \|x - y\|$ (X, d)

$\mathbb{R}^N$ Cerrado y acotado $\Leftrightarrow$ Para toda $(x_n) \in K^{\mathbb{N}}$, $\exists \varphi$, $(x_{\varphi(n)})_n$ que cv (en K).



B es acotado pero no compacto.

$x_n \to (\sqrt{2} - 1, 2)$

$x_{\varphi(n)} \to (\sqrt{2} - 1, 2) \notin B$

$(\mathbb{R}^{\mathbb{N}}, \|\cdot\|_\infty)$, $\sup_{n \in \mathbb{N}} |x_n|$

$B_c(0,1) = \{x \in \mathbb{R}^{\mathbb{N}} \mid \|x - 0\|_\infty \leq 1\}$

$a_n = (0, \ldots, \underset{\overset{\uparrow}{n}}{1}, 0, \ldots, 0)$